Prediction

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Predictions, fitted values, residuals, and prediction errors

Definition 0.1 (Prediction (predicted value))**.** A **prediction** (or **predicted value**) $\hat{y}(x)$ of an outcome Y with covariate value $X = x$ is a number computed from x and from data $(x_1, y_1), \ldots, (x_n, y_n)$ by a rule g, and used as a guess of Y:

$$\hat{y}(x) \stackrel{\text{def}}{=} g(x; (x_1, y_1), \ldots, (x_n, y_n)).$$

Example 0.1 (Predicting with an estimated conditional mean)**.** Suppose a model specifies the conditional mean $\mathrm{E}[Y \mid X = x]$ as a function $\mu(x; \underset{\sim}{\theta})$ of a parameter vector $\underset{\sim}{\theta}$, and $\underset{\sim}{\hat{\theta}}$ is an estimate of $\underset{\sim}{\theta}$. The usual prediction (Definition 0.1) of Y at $X = x$ is the estimated conditional mean:

$$\hat{y}(x) = \mu(x; \underset{\sim}{\hat{\theta}}).$$

Remark (Why the estimated conditional mean is the usual prediction)

Remark 0.1 (Why the estimated conditional mean is the usual prediction). The prediction in Example 0.1 is chosen because the conditional mean is the best possible prediction under squared error loss. Among all functions h of the covariate, $\mathrm{E}[Y \mid X]$ has the smallest expected squared error $\mathrm{E}\left[(Y - h(X))^2\right]$ (conditional mean minimizes squared-error risk). The conditional mean is unknown, so we estimate it from data and predict with the estimate.
The target depends on the loss. For absolute error loss, the best prediction is the conditional median instead, and other losses have other targets (Hastie et al. 2009, 20).

Definition 0.2 (Fitted value)**.** The **fitted value** $\hat{y}_i$ of observation i is the prediction at its own covariate value x_i:

$$\hat{y}_i \stackrel{\text{def}}{=} \hat{y}(x_i).$$

Remark (Why the fitted value is the estimated conditional mean)

Remark 0.2 (Why the fitted value is the estimated conditional mean). The fitted value is the prediction at the observed covariate value x_i, so for the prediction in Example 0.1 it is the estimated conditional mean at x_i:

$$\hat{y}_i = \mu(x_i; \underset{\sim}{\hat{\theta}}).$$

It is chosen for the reason in Remark 0.1: the conditional mean is the best prediction under squared error loss.

Definition 0.3 (Residual)**.** The **residual** for observation i is its observed outcome minus its fitted value:

$$r_i \stackrel{\text{def}}{=} y_i - \hat{y}_i$$

Definition 0.4 (Prediction error)**.** The **prediction error** of a prediction $\hat{y}$ (such as $\hat{y}(x)$ or a fitted value $\hat{y}_i$) of an observed outcome y is the prediction minus the outcome:

$$\varepsilon \stackrel{\text{def}}{=} \hat{y} - y.$$

Exercise 0.1 (Prediction error and residual)**.** For an observation i, write its prediction error ε_i in terms of its residual r_i.

Solution

Solution 0.1.

$$\begin{aligned} \varepsilon_i &= \hat{y}_i - y_i && \text{(definition of prediction error)} \\ &= -(y_i - \hat{y}_i) && \text{(factor out } -1) \\ &= -r_i && \text{(definition of residual)} \end{aligned}$$

Theorem 0.1 (Prediction error is the negative of the residual)**.** *For an observation i, the prediction error of its fitted value is the negative of its residual:*

$$\varepsilon_i = -r_i.$$

Proof

Proof. This is the solution to Exercise 0.1. □

Corollary 0.1 (The residual is the negative of the prediction error)**.** *For an observation i, its residual is the negative of the prediction error of its fitted value:*

$$r_i = -\varepsilon_i.$$

Proof

Proof. Multiply both sides of Theorem 0.1 by -1. □

Example 0.2 (Residuals and prediction errors for birth weights)**.** The `birthwt` data record the birth weight (grams) of 189 babies and their mothers' weight at the last menstrual period (pounds), from a study at Baystate Medical Center in Springfield, Massachusetts (Hosmer and Lemeshow 1989). They ship with the R package `MASS` (`MASS::birthwt`), and this site has a copy at `data/birthwt.csv`.
The code below fits a line predicting birth weight from mother's weight, then compares each residual with the negative of its prediction error.

R

```r
birthwt <- read.csv("data/birthwt.csv")
fit <- lm(bwt ~ lwt, data = birthwt)
fitted_bwt <- fitted(fit)
residual <- birthwt$bwt - fitted_bwt
prediction_error <- fitted_bwt - birthwt$bwt
head(data.frame(
  bwt = birthwt$bwt,
  fitted = round(fitted_bwt, 1),
  residual = round(residual, 1),
  minus_prediction_error = round(-prediction_error, 1)
))
#>    bwt fitted residual minus_prediction_error
#> 1 2523 3175.7   -652.7                 -652.7
#> 2 2551 3056.1   -505.1                 -505.1
#> 3 2557 2834.7   -277.7                 -277.7
#> 4 2594 2848.0   -254.0                 -254.0
#> 5 2600 2843.5   -243.5                 -243.5
#> 6 2622 2918.8   -296.8                 -296.8
max(abs(residual - (-prediction_error)))
#> [1] 0
```

Python

```python
import numpy as np
import pandas as pd

birthwt = pd.read_csv("data/birthwt.csv")
X = np.column_stack([np.ones(len(birthwt)), birthwt["lwt"]])
y = birthwt["bwt"].to_numpy(dtype=float)
coef = np.linalg.lstsq(X, y, rcond=None)[0]
fitted_bwt = X @ coef
residual = y - fitted_bwt
prediction_error = fitted_bwt - y
print(pd.DataFrame({
    "bwt": y,
    "fitted": fitted_bwt.round(1),
    "residual": residual.round(1),
    "minus_prediction_error": (-prediction_error).round(1),
}).head())
#>       bwt  fitted  residual  minus_prediction_error
#> 0  2523.0  3175.7    -652.7                  -652.7
#> 1  2551.0  3056.1    -505.1                  -505.1
#> 2  2557.0  2834.7    -277.7                  -277.7
#> 3  2594.0  2848.0    -254.0                  -254.0
#> 4  2600.0  2843.5    -243.5                  -243.5
print(np.max(np.abs(residual - (-prediction_error))))
#> 0.0
```

The last two columns agree in every row, and the largest difference between them over all 189 babies is 0, as Corollary 0.1 says.

Remark (Prediction errors and residuals)

Remark 0.3 (Prediction errors and residuals). The prediction in Definition 0.4 can be a fitted value, for an observation used to fit the model, or a prediction for an observation that was not used to fit the model. Prediction error is oriented the same way as estimation error: estimate minus true value.

$$\varepsilon = \hat{y} - y \quad \text{(prediction error)}$$
$$\varepsilon\left(\hat{\theta}\right) = \hat{\theta} - \theta \quad \text{(estimation error)}$$

For a fitted value, the prediction error is the negative of the residual (Theorem 0.1), so the two differ only in orientation.

Sample mean squared error of predictions

Definition 0.5 (Mean squared error of predictions)**.** The **sample mean squared error (sMSE)** of predictions, also called their **mean squared error**, $\hat{y}_1, \ldots, \hat{y}_n$ of observed outcomes $y_1, \ldots, y_n$ is the mean of their squared prediction errors:

$$\text{sMSE}(\hat{y}) \stackrel{\text{def}}{=} \frac{1}{n} \sum_{i=1}^{n} \varepsilon_i^2.$$

Remark (Prediction sMSE, estimator pMSE and RSS)

Remark 0.4 (Prediction sMSE, estimator pMSE and RSS). Definition 0.5 averages squared errors over observations, where the pMSE of an estimator takes an expectation over repeated samples. On the data used to fit the model, the sMSE of the fitted values is the residual sum of squares divided by n (sMSE and RSS):

$$\text{sMSE}(\hat{y}) = \frac{1}{n}\text{RSS}(\underset{\sim}{\hat{\theta}}), \qquad \text{RSS}(\underset{\sim}{\hat{\theta}}) = n \text{ sMSE}(\hat{y}).$$

Example 0.3 (Fitted values, residuals, and prediction errors for three points)**.** Take the points $(0, 1)$, $(1, 2)$, $(2, 2)$ and the line with $\beta_0 = 1$ and $\beta_x = 0.5$ from the residual sum of squares example, and treat the line as the fitted model.

x_i	y_i	$\hat{y}_i = 1 + 0.5x_i$	$r_i = y_i - \hat{y}_i$	$\varepsilon_i = \hat{y}_i - y_i$
0	1	1	0	0
1	2	1.5	0.5	−0.5
2	2	2	0	0

> Figure 1 shows the observed outcomes, the fitted values on the line, and the residual of the second point.

```r
fre_points <- tibble::tibble(x = c(0, 1, 2), y = c(1, 2, 2)) |>
  dplyr::mutate(fitted = 1 + 0.5 * x)

ggplot2::ggplot(fre_points, ggplot2::aes(x = x)) +
  ggplot2::geom_abline(intercept = 1, slope = 0.5) +
  ggplot2::geom_segment(
    ggplot2::aes(y = fitted, xend = x, yend = y),
    linetype = "dashed",
    color = "firebrick"
  ) +
  ggplot2::geom_point(ggplot2::aes(y = fitted), shape = 1, size = 3) +
  ggplot2::geom_point(ggplot2::aes(y = y), size = 3) +
  ggplot2::annotate(
    "text",
    x = 1.1, y = 1.75, hjust = 0,
    label = "r[2] == 0.5", parse = TRUE
  ) +
  ggplot2::labs(y = "y")
```

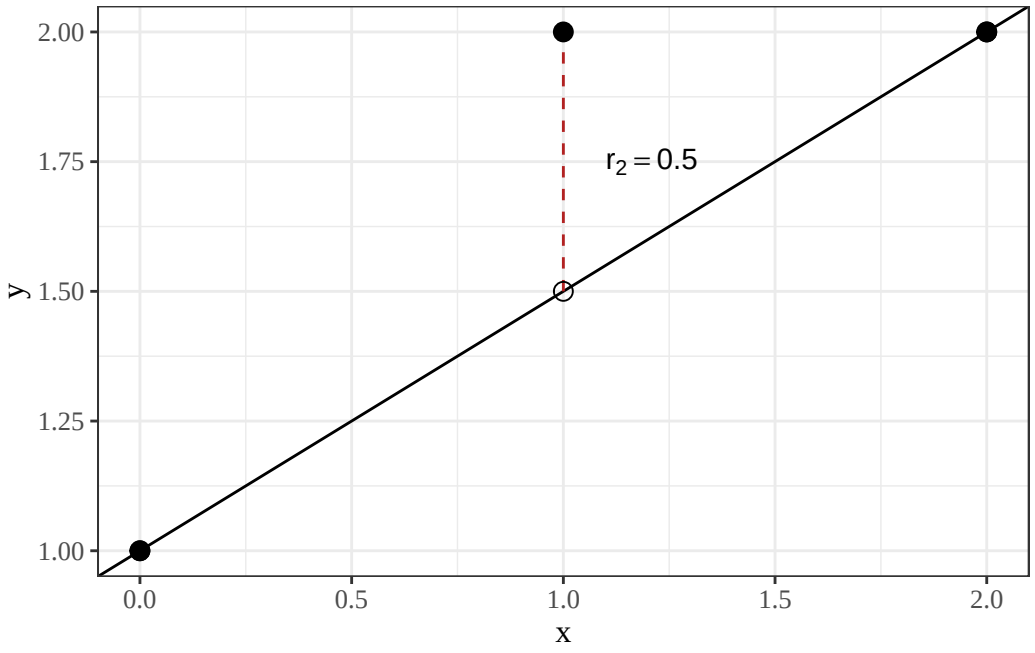


Figure 1: Observed outcomes (filled), fitted values on the line (open), and the residual of the second point (dashed), which points up from its fitted value to its observed outcome

The sMSE of these predictions is

$$\begin{aligned}\text{sMSE}(\hat{y}) &= \frac{0^2 + (-0.5)^2 + 0^2}{3} \\ &= \frac{0.25}{3} \\ &\approx 0.083,\end{aligned}$$

the residual sum of squares, 0.25, divided by $n = 3$.

Estimation versus prediction

Remark (Estimation versus prediction)

Remark 0.5 (Estimation versus prediction). Estimation and prediction both measure error as a signed difference, and both summarize it with a mean squared error (a pMSE for an estimator, an sMSE for predictions), but they aim at different targets:

- **Target.** An estimate aims at a fixed, unknown estimand. A prediction aims at an outcome y, which is a random quantity even when the model is known exactly.
- **Error.** The estimation error is the estimate minus the estimand. The prediction error is the prediction minus the outcome.
- **Mean squared error.** The pMSE of an estimator is an expectation over repeated samples. The sMSE of predictions is an average over observations.
- **What cannot be removed.** The error of a consistent estimator shrinks toward zero as the sample grows. A prediction's expected squared error cannot fall below the irreducible error, the variance of the outcome around its conditional mean.

Validating Predictions estimates the prediction error on new observations.

Hastie, Trevor, Robert Tibshirani, and Jerome Friedman. 2009. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. 2nd ed. Springer. https://doi.org/10.1007/978-0-387-84858-7.

Hosmer, David W., and Stanley Lemeshow. 1989. *Applied Logistic Regression*. Wiley.